

NEXUS Team Contribution Guide

Evidence-First Investigative Intelligence | Six-member GitHub workflow

Repository: github.com/Aryansh077/NEXUS-Investigative-Intelligence
Use synthetic data only. Never treat an anomaly as proof of criminality.

Team Assignments

Member 1: Data Ingestion and Backend

`feature/ingestion-backend`

Upload APIs, parsing, normalization, validation, database storage, and evidence registration.

Folders: `backend/app/services/ingestion/`,
`backend/app/api/ingestion.py`, `backend/app/database/`,
`scripts/seed_database.py`

Member 2: NLP and Entity Resolution

`feature/nlp-resolution`

Entity extraction, aliases, fuzzy matching, confidence scores, and relationship extraction.

Folders: `backend/app/services/nlp/`, `entity_resolution/`,
`relationship_extraction/`

Member 3: Knowledge Graph

`feature/knowledge-graph`

Graph construction, paths, centrality, communities, and Cypher artifacts.

Folders: `backend/app/services/graph/`, `neo4j/`

Member 4: ML and Anomaly Detection

`feature/ml-anomaly`

Feature engineering, Isolation Forest, anomaly explanations, evaluation, and ML tests.

Folders: `ml/`, `backend/app/services/anomaly/`

Member 5: Frontend and Visualization

`feature/frontend-visualization`

Dashboard, entities, graph, timeline, evidence, anomalies, Copilot UI, and API states.

Folder: `frontend/`

Member 6: Copilot, Security, and Integration

`feature/copilot-security`

Grounded Copilot, authentication, RBAC, audit, Docker, CI, and integration checks.

Folders: `backend/app/services/copilot/`, `backend/app/security/`,
`Dockerfile`, `.github/`

First-time Setup

```
git clone https://github.com/Aryansh077/NEXUS-Investigative-Intelligence.git
Set-Location NEXUS-Investigative-Intelligence
git switch YOUR_ASSIGNED_BRANCH
git pull origin YOUR_ASSIGNED_BRANCH
```

Run and Test

```
Backend terminal:
Set-Location backend
python -m venv .venv
.\venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python ..\scripts\generate_data.py
python ..\scripts\seed_database.py
python -m uvicorn app.main:app --reload --port 8000
```

```
Frontend terminal:
Set-Location frontend
npm install
npm run dev -- --host 0.0.0.0 --port 5173
```

Open <http://localhost:5173>. API docs: <http://localhost:8000/docs>. Demo login: investigator / nexus-demo.

Before a Pull Request

```
Set-Location backend
.\.venv\Scripts\python.exe -m pytest -q
Set-Location ..\frontend
npm run build
Set-Location ..
git add .
git commit -m "Describe the focused change"
git push -u origin YOUR_ASSIGNED_BRANCH
```

Open a Pull Request with base branch main. Another member should review it before merge.

Team Rules

- Do not develop directly on main.
- Do not commit .env, .venv, node_modules, database files, credentials, or private data.
- Use synthetic demonstration data only.
- Update tests when backend behavior changes.
- Do not claim model accuracy or training results that were not measured.
- Anomaly results are investigative leads requiring human verification.

NEXUS | From fragmented data to connected intelligence | Local SIH prototype